

1. COVER PAGE:

CPSC 304 Project Cover Page

Milestone #: 2

Date: March 3rd, 2025

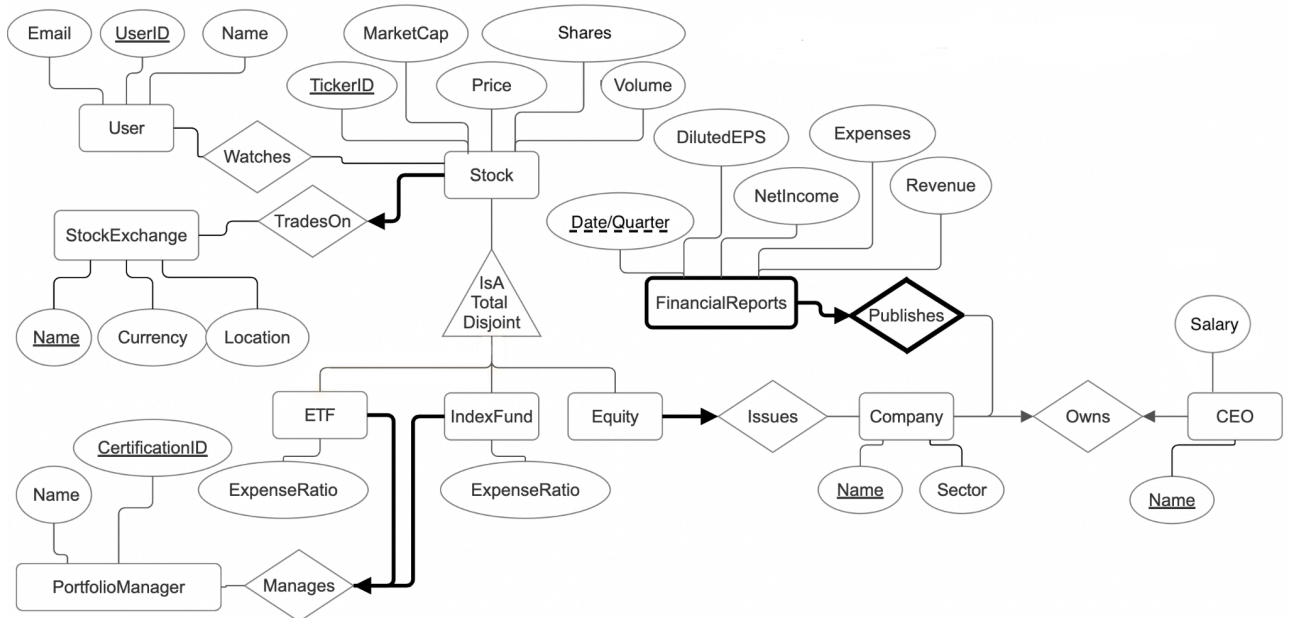
Group Number: 8

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. The goal of the application is to allow users to have easy access to important information relevant to a particular stock or index fund. The database will allow users to view stocks, index funds, or ETFs that are being traded on a sample market. The database will have up to date information on these stocks including trading value, volume, market cap, relevant news, and financial reports.
3. Changes made to ER diagram include:
 - a. using a dotted line for partial key to follow in-class notation style
 - b. salary key was added for the CEO entity to fulfil requirements for entities
 - c. “52WeekHigh/Low” attribute was replaced with the “Shares” attribute for the Stock entity to include a valid FD.



4. Schemas:
 - a. StockWithExchangeRelation (**TickerID**: CHAR[5], MarketCap: DECIMAL(n, 2), Price: DECIMAL(n, 2), Shares: INTEGER, Volume: INTEGER, **Name**: VARCHAR)
 - i. PK: TickerID
 - ii. FK: Name
 - iii. Constraint: Name is not NULL
 - b. EquityStockIssuesCompany (**TickerID**: CHAR[5], **Name**: VARCHAR)
 - i. PK: TickerID
 - ii. FK: TickerID, Name
 - iii. Constraint: Name is not NULL
 - c. IndexFundManagesPM (**TickerID**: CHAR[5], ExpenseRatio: DECIMAL(n,2), **CertificationID**: INTEGER)
 - i. PK: TickerID
 - ii. FK: TickerID, CertificationID
 - iii. Constraint: CertificationID is not NULL
 - d. ETFManagesPM(**TickerID**: CHAR[5], ExpenseRatio: DECIMAL(n,2), **CertificationID**: VARCHAR)
 - i. PK: TickerID
 - ii. FK: TickerID, CertificationID
 - iii. Constraint: CertificationID is not NULL
 - e. StockExchange (**Name**: VARCHAR, Currency: VARCHAR, Location: VARCHAR)
 - i. PK: Name
 - f. User (**UserID**: VARCHAR, Email: VARCHAR, Name: VARCHAR)
 - i. PK: UserID
 - ii. CK: UserID, Email

- g. UserWatchesStockRelation (UserID: VARCHAR, TickerID: CHAR[5])
 - i. PK: UserID and TickerID
 - ii. FK: UserID, TickerID
- h. CEOOwnsCompany(Name: VARCHAR, Salary: DECIMAL(n,2), **Company.Name**: VARCHAR)
 - i. PK: Name
 - ii. FK: Company.Name
 - iii. Constraint: Company.Name is not NULL and UNIQUE
- i. Company(Name: VARCHAR, Sector: VARCHAR)
 - i. PK: Name
- j. FinancialReports(Date/Quarter: VARCHAR, DilutedEPS: DECIMAL(n,2), NetIncome: DECIMAL(n,2), Expenses: DECIMAL(n,2), Revenue: DECIMAL(n,2), **Name**: VARCHAR)
 - i. PK: Date/Quarter and Name
 - ii. FK: Name

5. Functional Dependencies:

- a. StockWithExchangeRelation(TickerID, MarketCap, Price, Shares, Volume, **Name**)
 - i. FD1 (PK): TickerID $\rightarrow$ Price, Shares, Volume, Name
 - ii. FD2: Price, Shares $\rightarrow$ MarketCap
- b. EquityStockIssuesCompany (TickerID, **Name**)
 - i. FD (PK): TickerID $\rightarrow$ Name
- c. IndexFundManagesPM (TickerID, ExpenseRatio, **CertificationID**)
 - i. FD(PK): TickerID $\rightarrow$ ExpenseRatio, CertificationID
- d. ETFManagesPM (TickerID, ExpenseRatio, **CertificationID**)
 - i. FD(PK): TickerID $\rightarrow$ ExpenseRatio, CertificationID
- e. StockExchange (Name, Currency, Location)
 - i. FD1(PK): Name $\rightarrow$ Location
 - ii. FD2: Location $\rightarrow$ Currency
- f. User (UserID, Email, Name)
 - i. FD1(PK): UserID $\rightarrow$ Email, Name
 - ii. FD2(CK): Email $\rightarrow$ UserID, Name
- g. UserWatchesStockRelation (UserID, TickerID)
 - i. No FDs
- h. CEOOwnsCompany (Name, Salary, **Company.Name**)
 - i. FD1(PK): Name $\rightarrow$ Salary, CompanyName
 - ii. FD2: CompanyName $\rightarrow$ Name
- i. Company (Name, Sector)
 - i. FD1 (PK): Name $\rightarrow$ Sector
- j. FinancialReports (Date/Quarter, DilutedEPS, NetIncome, Expenses, Revenue, **Name**)
 - i. FD1(PK): Date\Quarter, Name $\rightarrow$ DiluteEPS, Expenses, Revenue
 - ii. FD2: Revenue, Expenses $\rightarrow$ NetIncome

6. Normalization to BCNF

- a. StockWithExchangeRelation (TickerID: CHAR[5], MarketCap: DECIMAL(n, 2), Price: DECIMAL(n, 2), Shares: INTEGER, Volume: INTEGER, **Name**: VARCHAR)
 - i. Not in BCNF because FD2 is not a key. Decompose into the following:
 - 1. StockWithExchangeRelation (TickerID: CHAR[5], Price: DECIMAL(n, 2), Shares: INTEGER, Volume: INTEGER, **Name**: VARCHAR)
 - 2. PriceWithShares(Price: DECIMAL(n, 2), Shares: INTEGER, MarketCap: DECIMAL(n,2))
- b. EquityStockIssuesCompany (TickerID: CHAR[5], **Name**: VARCHAR)
 - i. Already in BCNF: closure of TickerID includes all attributes.

- c. IndexFundManagesPM (**TickerID**: CHAR[5], ExpenseRatio: DECIMAL(n,2), **CertificationID**: INTEGER)
 - i. Already in BCNF: closure of TickerID includes all attributes.
- d. ETFManagesPM (**TickerID**: CHAR[5], ExpenseRatio: DECIMAL(n,2), **CertificationID**: INTEGER)
 - i. Already in BCNF: closure of TickerID includes all attributes.
- e. StockExchange (**Name**: VARCHAR, Currency: VARCHAR, Location: VARCHAR)
 - i. Not in BCNF because FD2 is not a key. Decompose into the following:
 1. StockExchangeNameLocation (**Name**: VARCHAR, Location: VARCHAR)
 2. StockExchangeLocationCurrency (**Location**: VARCHAR, Currency: VARCHAR)
- f. User (**UserID**: VARCHAR, Email: VARCHAR, Name: VARCHAR)
 - i. Already in BCNF: closure of UserID includes all attributes; closure of Email includes all attributes.
- g. UserWatchesStockRelation (**UserID**: VARCHAR, **TickerID**: CHAR[5])
 - i. No FDs
- h. CEOOwnsCompany (**Name**: VARCHAR, Salary: DECIMAL(n,2), **Company.Name**: VARCHAR)
 - i. Already in BCNF: closure of Name includes all attributes, closure of Company.Name includes all attributes.
- i. Company (**Name**: VARCHAR, Sector: VARCHAR)
 - i. Already in BCNF: closure of Name includes all attributes.
- j. FinancialReports (**Date/Quarter**: VARCHAR, DilutedEPS: DECIMAL(n,2), NetIncome: DECIMAL(n,2), Expenses: DECIMAL(n,2), Revenue: DECIMAL(n,2), **Name**: VARCHAR)
 - i. Not in BCNF because FD2 is not a key. Decompose into the following:
 1. FinancialReport (**Date/Quarter**: VARCHAR, DilutedEPS: DECIMAL(n,2), Expenses: DECIMAL(n,2), Revenue: DECIMAL(n,2), **Name**: VARCHAR)
 2. RevenueExpensesNetIncome (Expenses: DECIMAL(n,2), Revenue: DECIMAL(n,2), NetIncome: DECIMAL(n,2))

7. SQL DDL statements:

a. StockWithExchangeRelation

```
CREATE TABLE StockWithExchangeRelation (
    TickerID    CHAR[5]          NOT NULL,
    Price       DECIMAL(n, 2),
    Shares      INTEGER,
    Volume      INTEGER,
    Name        VARCHAR          NOT NULL,
    PRIMARY KEY (TickerID),
    FOREIGN KEY (Name)
        REFERENCES StockExchange(Name)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

- b. CREATE TABLE PriceWithShares (
 Price DECIMAL(n, 2) NOT NULL,
 Shares INTEGER NOT NULL,
 MarketCap DECIMAL(n,2) ,
 PRIMARY KEY (Price, Shares)
);
- c. CREATE TABLE EquityStockIssuesCompany(
 Name VARCHAR NOT NULL,
 TickerID CHAR[5] NOT NULL,
 PRIMARY KEY (Name, TickerID),
 FOREIGN KEY (TickerID)
 REFERENCES StockWithExchangeRelation(TickerID)
 ON DELETE CASCADE
 ON UPDATE CASCADE
);
- d. CREATE TABLE User(
 UserID VARCHAR NOT NULL,
 Email VARCHAR,
 Name VARCHAR NOT NULL,
 PRIMARY KEY(UserID)
);
- e. CREATE TABLE IndexFundManagesPM(
 TickerID CHAR[5],
 ExpenseRatio DECIMAL(n,2) NOT NULL,
 CertificateID INTEGER
 PRIMARY KEY (TickerID),
 FOREIGN KEY (CertificateID)
 REFERENCES PortfolioManger(CertificateID),
 FOREIGN KEY (TickerID)
 REFERENCES StockWithExchangeRelation(TickerID)
 ON DELETE CASCADE
 ON UPDATE CASCADE
);
- f. CREATE TABLE StockExchange (
 Name VARCHAR,
 Location VARCHAR,
 PRIMARY KEY (Name),
 FOREIGN KEY (Location)
 REFERENCES LocationAndCurrency(TickerID)
 ON DELETE CASCADE
 ON UPDATE CASCADE
);
- g. CREATE TABLE LocationAndCurrency (
 Location VARCHAR,
 Currency VARCHAR NOT NULL,
 PRIMARY KEY (Location)
);

- h. CREATE TABLE ETFManagesPM (
 TickerID CHAR[5],
 ExpenseRatio DECIMAL(n,2) NOT NULL,
 CertificateID INTEGER,
 PRIMARY KEY (TickerID),
 FOREIGN KEY(CertificateID)
 REFERENCES PortfolioManager(CertificateID),
 FOREIGN KEY (TickerID)
 REFERENCES StockWithExchangeRelation(TickerID)
 ON DELETE CASCADE
 ON UPDATE CASCADE
);
- i. CREATE TABLE FinancialReport (
 Name CHAR[5],
 Date/Quarter VARCHAR,
 DilutedEPS DECIMAL(n,2) NOT NULL,
 Expenses DECIMAL(n,2) NOT NULL,
 Revenue DECIMAL(n,2) NOT NULL,
 PRIMARY KEY (Name, Date/Quarter)
);
- j. CREATE TABLE RevenueAndExpenses(
 Revenue DECIMAL(n,2),
 Expenses DECIMAL(n,2),
 NetIncome DECIMAL(n,2) NOT NULL,
 PRIMARY KEY (Revenue, Expenses)
);
- k. CREATE TABLE CEOOwnsCompany (
 CompanyName VARCHAR,
 CEOName VARCHAR NOT NULL,
 PRIMARY KEY (CompanyName, CEOName)
 UNIQUE(CompanyName)
 FOREIGN KEY (CompanyName)
 REFERENCES Company(CompanyName),
 ON DELETE CASCADE
 ON UPDATE CASCADE,
);
- l. CREATE TABLE Company (
 CompanyName VARCHAR,
 Sector VARCHAR NOT NULL,
 PRIMARY KEY (CompanyName),
);

```

m. CREATE TABLE UserWatchesStockRelation(
    UserID      VARCHAR,
    TickerID     char[5],
    PRIMARY KEY(UserID, TickerID),
    FOREIGN KEY(UserID)
        REFERENCES User(UserID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    FOREIGN KEY(TickerID)
        REFERENCES Stock(TickerID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

8. INSERT statements:

a. StockWithExchangeRelation:

- i. INSERT INTO StockWithExchangeRelation (TickerID, Price, Shares, Volume, Name) VALUES ('temp1', 900.00, 10000, 99000, 'tempCompany1');
- ii. INSERT INTO StockWithExchangeRelation (TickerID, Price, Shares, Volume, Name) VALUES ('temp2', 800.00, 10000, 88000, 'tempCompany2');
- iii. INSERT INTO StockWithExchangeRelation (TickerID, Price, Shares, Volume, Name) VALUES ('temp3', 700.00, 10000, 77777, 'tempCompany3');
- iv. INSERT INTO StockWithExchangeRelation (TickerID, Price, Shares, Volume, Name) VALUES ('temp4', 600.00, 10000, 66666, 'tempCompany4');
- v. INSERT INTO StockWithExchangeRelation (TickerID, Price, Shares, Volume, Name) VALUES ('temp5', 500.00, 10000, 55555, 'tempCompany5');

b. PriceWithShares:

- i. INSERT INTO PriceWithShares (Price, Shares, MarketCap) VALUES (900.00, 10000, 9000000.00,);
- ii. INSERT INTO PriceWithShares (Price, Shares, MarketCap) VALUES (800.00, 10000, 8000000.00,);
- iii. INSERT INTO PriceWithShares (Price, Shares, MarketCap) VALUES (700.00, 10000, 7000000.00,);
- iv. INSERT INTO PriceWithShares (Price, Shares, MarketCap) VALUES (600.00, 10000, 6000000.00,);
- v. INSERT INTO PriceWithShares (Price, Shares, MarketCap) VALUES (500.00, 10000, 5000000.00,);

c. EquityStockIssuesCompany:

- i. INSERT INTO EquityStockIssuesCompany (Name, TickerID) VALUES ('tempCompany1', 'temp1');
- ii. INSERT INTO EquityStockIssuesCompany (Name, TickerID) VALUES ('tempCompany2', 'temp2');
- iii. INSERT INTO EquityStockIssuesCompany (Name, TickerID) VALUES ('tempCompany3', 'temp3');
- iv. INSERT INTO EquityStockIssuesCompany (Name, TickerID) VALUES ('tempCompany4', 'temp4');
- v. INSERT INTO EquityStockIssuesCompany (Name, TickerID) VALUES ('tempCompany5', 'temp5');

d. StockExchange:

- i. INSERT INTO StockExchange(Name, Location) VALUES ('tempCompany1','Random Place 1');
- ii. INSERT INTO StockExchange(Name, Location) VALUES ('tempCompany2','Random Place 2');
- iii. INSERT INTO StockExchange(Name, Location) VALUES ('tempCompany3','Random Place 3');
- iv. INSERT INTO StockExchange(Name, Location) VALUES ('tempCompany4','Random Place 4');
- v. INSERT INTO StockExchange(Name, Location) VALUES ('tempCompany5','Random Place 5');

e. LocationAndCurrency:

- i. INSERT INTO LocationAndCurrency(Location,Currency) VALUES ('Random Place 1','tempC1');
- ii. INSERT INTO LocationAndCurrency(Location,Currency) VALUES ('Random Place 2','tempC2');
- iii. INSERT INTO LocationAndCurrency(Location,Currency) VALUES ('Random Place 3','tempC3');
- iv. INSERT INTO LocationAndCurrency(Location,Currency) VALUES ('Random Place 4','tempC4');
- v. INSERT INTO LocationAndCurrency(Location,Currency) VALUES ('Random Place 5','tempC5');

f. ETFManagesPM:

- i. INSERT INTO ETFManagesPM(TickerID,ExpenseRatio, CertificateID) Values ('temp6', 0.01, 1);
- ii. INSERT INTO ETFManagesPM(TickerID,ExpenseRatio, CertificateID) Values ('temp7',0.1, 2);
- iii. INSERT INTO ETFManagesPM(TickerID,ExpenseRatio, CertificateID) Values ('temp8',0.3, 3);
- iv. INSERT INTO ETFManagesPM(TickerID,ExpenseRatio, CertificateID) Values ('temp9',0.5, 4);
- v. INSERT INTO ETFManagesPM(TickerID,ExpenseRatio, CertificateID) Values ('temp10',0.7, 5);

g. User:

- i. INSERT INTO User(UserID, Email, Name) VALUES('tempUID1', 'tempU1@email.com', 'tempUser1');
- ii. INSERT INTO User(UserID, Email, Name) VALUES('tempUID2', 'tempU2@email.com', 'tempUser2');
- iii. INSERT INTO User(UserID, Email, Name) VALUES('tempUID3', 'tempU3@email.com', 'tempUser3');
- iv. INSERT INTO User(UserID, Email, Name) VALUES('tempUID4', 'tempU4@email.com', 'tempUser4');
- v. INSERT INTO User(UserID, Email, Name) VALUES('tempUID5', 'tempU5@email.com', 'tempUser5');

h. FinancialReport:

- i. INSERT INTO FinancialReport (Name, DateQuarter, DilutedEPS, Expenses, Revenue) VALUES ('tempCompany1','2000Q1',4.8, 5000, 150000);
- ii. INSERT INTO FinancialReport (Name, DateQuarter, DilutedEPS, Expenses, Revenue) VALUES ('tempCompany2','2002Q3',2.0, 2444, 5000000);
- iii. INSERT INTO FinancialReport (Name, DateQuarter, DilutedEPS, Expenses, Revenue) VALUES ('tempCompany3','2005Q2'1.9, 4213, 542341);
- iv. INSERT INTO FinancialReport (Name, DateQuarter, DilutedEPS, Expenses, Revenue) VALUES ('tempCompany4','2013Q4',0.75, 10000, 999999);
- v. INSERT INTO FinancialReport (Name, DateQuarter, DilutedEPS, Expenses, Revenue) VALUES ('tempCompany5','2016Q3',3.2, 400, 152333);

i. IndexFundManagesPM:

- i. INSERT INTO IndexFundManagesPM (TickerID, ExpenseRatio, CertificateID) VALUES('temp1', 0.01, 1);
- ii. INSERT INTO IndexFundManagesPM (TickerID, ExpenseRatio, CertificateID) VALUES('temp2', 0.02, 2);
- iii. INSERT INTO IndexFundManagesPM (TickerID, ExpenseRatio, CertificateID) VALUES('temp3', 0.02, 3);
- iv. INSERT INTO IndexFundManagesPM (TickerID, ExpenseRatio, CertificateID) VALUES('temp4', 0.05, 1);
- v. INSERT INTO IndexFundManagesPM (TickerID, ExpenseRatio, CertificateID) VALUES('temp5', 0.03,' 2);

j. RevenueAndExpenses:

- i. INSERT INTO RevenueAndExpenses (Revenue, Expenses, NetIncome) VALUES (150000.00, 50000.00, 100000.00);
- ii. INSERT INTO RevenueAndExpenses (Revenue, Expenses, NetIncome) VALUES (50000.00,10000.00,40000.00);
- iii. INSERT INTO RevenueAndExpenses (Revenue, Expenses, NetIncome) VALUES (7000.00,500.00,6500.00);
- iv. INSERT INTO RevenueAndExpenses (Revenue, Expenses, NetIncome) VALUES (900000.00,400000.00,500000.00);
- v. INSERT INTO RevenueAndExpenses (Revenue, Expenses, NetIncome) VALUES (200000.00,60000.00,140000.00);

k. CEOOwnsCompany:

- i. INSERT INTO CEOOwnsCompany(CompanyName, CEOName) VALUES('tempCompany1','tempCEO5');
- ii. INSERT INTO CEOOwnsCompany(CompanyName, CEOName) VALUES('tempCompany2','tempCEO2');
- iii. INSERT INTO CEOOwnsCompany(CompanyName, CEOName) VALUES('tempCompany3','tempCEO4');
- iv. INSERT INTO CEOOwnsCompany(CompanyName, CEOName) VALUES('tempCompany4','tempCEO1');
- v. INSERT INTO CEOOwnsCompany(CompanyName, CEOName) VALUES('tempCompany5','tempCEO3');

- I. Company:
 - i. INSERT INTO Company (CompanyName, Sector) VALUES ('tempCompany1', 'technology');
 - ii. INSERT INTO Company (CompanyName, Sector) VALUES ('tempCompany2', 'financial');
 - iii. INSERT INTO Company (CompanyName, Sector) VALUES ('tempCompany3', 'energy');
 - iv. INSERT INTO Company (CompanyName, Sector) VALUES ('tempCompany4', 'utilities');
 - v. INSERT INTO Company (CompanyName, Sector) VALUES ('tempCompany5', 'healthcare');
- m. UserWatchesStockRelation:
 - i. INSERT INTO UserWatchesStockRelation(UserID, TickerID) VALUES ('tempUID5', 'temp1');
 - ii. INSERT INTO UserWatchesStockRelation(UserID, TickerID) VALUES ('tempUID5', 'temp2');
 - iii. INSERT INTO UserWatchesStockRelation(UserID, TickerID) VALUES ('tempUID3', 'temp1');
 - iv. INSERT INTO UserWatchesStockRelation(UserID, TickerID) VALUES ('tempUID4', 'temp3');
 - v. INSERT INTO UserWatchesStockRelation(UserID, TickerID) VALUES ('tempUID1', 'temp5');

9. AI Acknowledgement: No AI was used in completing this milestone.